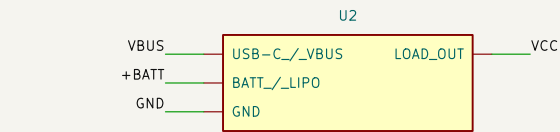


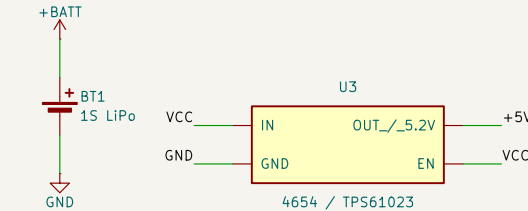
NFC—dismiss alarm clock | Breadboard prototype

Revision A — Module terminal names follow board silkscreen; onboard circuits and unused headers omitted.

USB / BATTERY / POWER

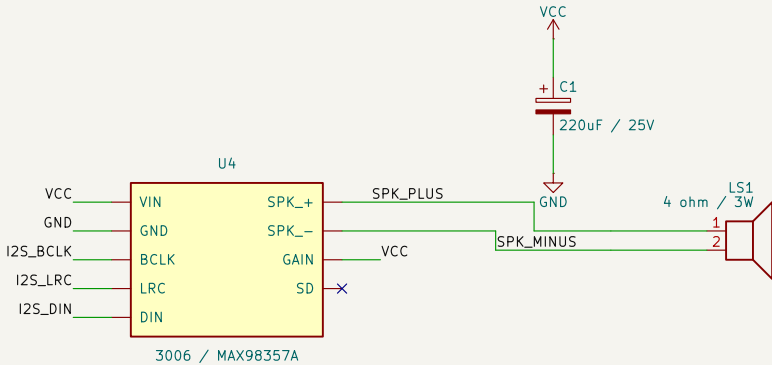


U2: built-in USB-C input, 5 V supply.
Set charge jumper to 500 mA for BT1.
VCC = LOAD OUT, approximately 3–4.4 V.



BT1: protected 1S LiPo, 4.2 V full. Check JST polarity.
U3: +5V net is nominally 5.2 V. EN tied to IN.
Disconnect U3 OUT from U1 5V before programming by USB.

I2S AUDIO



GAIN to VIN = 6 dB. SD open uses onboard mono mix.
Place C1 close to amplifier VIN/GND; observe polarity.
Keep speaker pair short; neither speaker wire goes to GND.

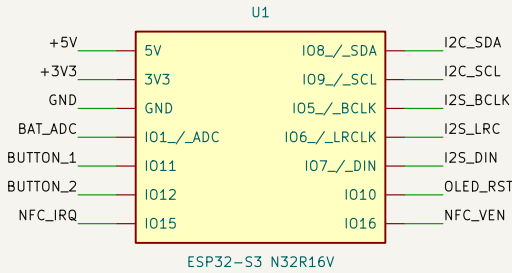
ASSEMBLY NOTES

All grounds common. I2C uses 3.3 V; begin at 100 kHz with short wires.
Breakout boards include pull-ups and decoupling; extra bus pull-ups are not fitted.
Test combined 3.3 V load during Wi-Fi + NFC activity before relying on battery mode.
Functional breadboard wiring only; no PCB layout or fabrication footprints.
U9 is the proposed PN7160 module; confirm this model before buying/wiring.

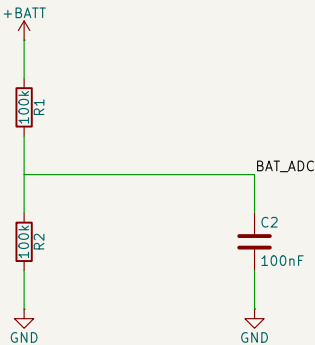
External USB / battery / ground source declarations:



WI-FI CONTROLLER

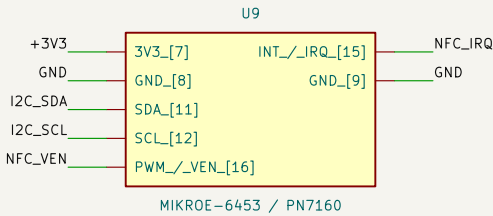


3.3 V GPIO. Configure octal flash and PSRAM.
Web configuration over Wi-Fi; MCU stays awake.



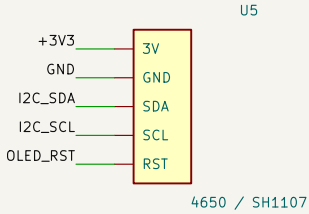
Battery divider: read BAT, not LOAD OUT.
ADC maximum 2.1 V; configure attenuation and calibrate. C2 filters the ADC input.

NFC READER



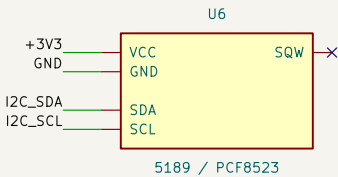
U9: MIKROE-6453 NFC 7 Click I2C / PN7160.
Set VCC_SEL to 3.3 V and interface straps to I2C.
VEN is mikroBUS PWM; IRQ is INT. RST is ClickID, unused.
Keep antenna clear of metal, speaker, and power wiring.

DISPLAY



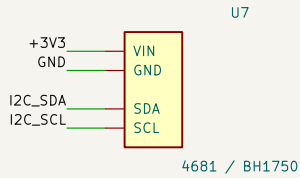
SH1107, I2C 0x3C. Pulse RST at startup.
Use Adafruit SH110X driver. OLED buttons unused.

REAL-TIME CLOCK



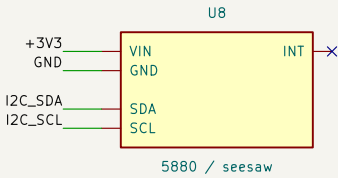
PCF8523, I2C 0x68. Insert CR1220 in onboard holder.
SQW unused: alarms handled by running firmware.

AMBIENT LIGHT



BH1750, default I2C 0x23. Power from 3.3 V.

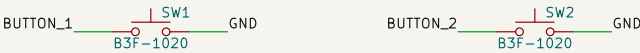
ROTARY ENCODER + PUSH



Seesaw, default I2C 0x36. Poll rotation and push over I2C.
INT unused. Module handles encoder decoding.

BUTTONS

Enable internal pull-ups; pressed = LOW.



Sheet: /
File: NFC_Alarm_Prototype.kicad_sch

Title:

Size: A3

Date:

Rev:

KiCad E.D.A. 10.0.6

Id: 1/1